# Opti-Hawk Meter


## Team 10219

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# Abstract


Among the eleven grand challenges that hamper Egypt's development progress, the two grand challenges—improving the scientific and technological environment and recycling garbage and waste for economic and environmental purposes—are addressed. Aligning with those challenges, the purpose of this study is to solve the problem of continuous inaccurate measurements of length variations regarding STEM students' experiments in the mechanics and physics lab. Hence, what was done is a networked laser meter relying on an infrared laser module and a direct Time-of-Flight (ToF) mechanism to measure the minute variations effectively. The prototype was tested to find out if it achieves the design requirements, which include a 5% error limit and a volume of no more than 0.002$m^2$ to guarantee portability. Moreover, data must be fully analyzed in graphs within 5 seconds at maximum from the measurements. After testing, the major findings observed that the error of the average of ten measurements is 2.7% compared with the standard ultrasonic sensor. In addition, graphs were generated in 3 seconds less than the required 5 seconds. To deduce, the project proved its applicability and reliability in the resuscitation of the electronic waste consumption; it transformed e-waste into a proficient solution for a real STEM community ethics concern.


# Introduction

Having been in the development generation, multiple challenges hinder Egypt's progress. The two main challenges are: improving the scientific and technological environment, and recycling garbage and waste for economic and environmental purposes. In fact, Egypt's GDP (Gross Domestic Product) –spent on technological applications– reached only 5.8% of the total GDP ***(Ilida, 2023).*** Furthermore, the number of imports in Egypt is about 82.5% more than the exports. Thereby, most money is wasted on buying imports, not recycling. Consequently, a rising number of 692 kilotons of electronic waste (e-waste) was produced in 2022.

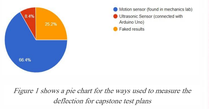



*Figure 1 shows a pie chart for the ways used to measure the deflection for capstone test plans*

The problem to be solved is the continuously inaccurate measurements of length variations regarding STEM students' experiments in the school's mechanics and physics labs. To illustrate, there are six main experiments in the curriculum for the 10th and 11th graders that focus on distance evaluation. Besides that, a project's test plan is often conducted relying mainly on the millimeter-scale distance variations, leading to the arising of ethical issues. For instance, 10$^{th}$ graders had to measure the deflection of their truss bridge in the previous academic year. As shown in **Figure 1**, 25.2% of the members fabricated the results before the exhibition. Although 66.4% measured deflection through the school's motion sensor, they didn't have any backup plan to ensure their results. Additionally, 81.2% of all teams are not sure about their results. Thus, some already tried solutions were considered to utilize their pros, avoiding their cons.

The first solution is a micro-centrifuge using a computer DVD drive. It is notorious for its low cost and the use of old DVD players. On the other hand, it has a limited range of 4 micro-tubes to carry at maximum. The second solution is the Bosch GLM 20 laser meter. It specializes in its high accuracy and effective timing, and a durable battery of 500 units. However, it relies only on the attached LCD screen to the device without formulating scientific graphs.

Some design requirements must be met to withstand the drawbacks of the prior solutions. Firstly, the error detected with distance measurements should not exceed 5% calibrated to an accredited device. Secondly, the prototype's overall volume should be less than 0.002 m$^3$ to enhance the feature of portability. Thirdly, a graph of measurements should be generated 5 seconds at maximum after the measurement.

Therefore, the chosen solution is an IoT (Internet of Things) laser meter to measure the minute variations on the milli-meter scale. Its basic idea relies on an infrared laser module emitting and receiving photons while pointing with a laser diode to calculate distances with the time-of-flight mechanism (ToF). In addition to the LCD screen displaying, Measurements could be transmitted through the Bluetooth module to a device, generating graphs through quick data analysis. The prototype is made of completely recycled e-waste. This will be discussed in detail in the following section.

# Materials

The project is totally made from efficient materials obeying all the design requirements, as illustrated in the following table:

| Name | Description | Old Device | Image |
|---|---|---|---|
| **ATmega8A** | An AVR microcontroller that functions as the brain of the whole device, calculating the distances | | |
| **Laser Module VL53L0X** | An infrared laser sensor emit invisible beams and then receive the reflected beams; signalling at their detection | | |
| **Laser diode** | A red pointing laser diode, which emit visible laser as an aiming mechanism for the meter | | |
| **LCD screen** | A displaying screen to show and demonstrate all the calculated measurements | **Unknown Old project** ——— | |
| **3.7-volt lithium battery** | A rechargeable battery acting as the power provider for the instrument | | |
| **Acrylic sheet** | A transparent 2.5 cm thickness acrylic sheet—crafted to be the rigid external structure of the device | ——— | |
| **Switch button** | A switch on/off for the device with 2 pins | | |
| **HC-05 Bluetooth Module** | A component responsible for the Bluetooth data transmitting between the device and the laptop | | |
| **Photovoltaic cells** | Three photovoltaic 1x2.5 cm cells stabilized on the device—a renewable power source besides the battery. | | |

# Methods

To visualize the recycled components assembly in the Opti-Hawk meter prototype, a 3D simulation for the prototype is created as shown in **Figure 2**. Afterward, the construction of the prototype is initiated by going through the following steps:

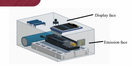



*Figure 2 shows Opti-Hawk meter prototype 3D design*

**Programming the circuit**

1- Using Cirkit Designer IDE simulator to create the whole circuit diagram, as demonstrated in **Figure 3.** This schematic diagram displays all the pins connections for all components; thus, preparing and planning for the circuit programming.

2- By using Arduino IDE application, the necessary libraries are downloaded –for all the specialized components– which are defined as pre-written software packages that contain functions and communication protocols used to control particular electronic components in the circuit.

3- Referencing the circuit diagram, three crucial libraries had to be imported: VL53L0X library, Liquid Crystall Display (LCD) needed library, and Software serial library—responsible for Bluetooth module functionality. Subsequently, these libraries are added into the code.

4- Continuously on Arduino IDE, the programming and implementation of the prototype's circuit code is done. In addition to libraries, completing the other normal circuit components' (without required libraries) functions within the code, like the laser diode, and battery. Hence, by taking pins connections in ***(Figure 2)*** into consideration, the organization for the code sequence is performed, accomplishing the connections between the meter components.



*Figure 3 shows the schematic diagram for the circuit*

5- After getting instructions from the Technology teacher, the final code for the circuit is ensured. **Figure 4** shows the conclusive code, ready to be uploaded to the ATmega8A—the central microprocessor of the device. The code is uploaded to the microcontroller via a USB-to-Serial converter cable.



*Figure 4 shows the code uploaded on the ATmega8A*

**Physical connection of the circuit**

1- The circuit is created and connected by using the breadboard. As shown in **Figure 5**, which includes the initial circuit creation controlled by the ATmega8A that is installed on the breadboard. In fact, the Arduino is only used as a code uploader to the ATmega8A in this early circuit.




*Figure 5 shows placement of the meter's components, completing the circuit.*

2-During the formation of this early circuit, at first, the components attached directly to breadboard are placed on it, which are: the ATmega8A, and the Bluetooth module linked to the ATmega8A through PB0(D8) , and PB1(D9) pins shown in **Figure 6**.

3-The other components (not attached to the breadboard) are wired to the breadboard, therefore a connection with the microprocessor. These components include: the laser diode wired to PD5(D5) pin, the laser module (VL53L0X) with PC5, PC4 pins –these are for the emission and calculation part– as well as the LCD (responsible for the measurements display).



*Figure 6 shows an illustration for the pinout and in of the ATmega8A*

4- For the power supply, the lithium battery is connected to a step-up module to increase the voltage of the battery to 5 volts. Then, three photovoltaic cells - connected in series - are attached to a battery charging module to charge the lithium battery. The two resulting wires from the charging module are connected to a voltage regulator to ensure stable voltage in the circuit.

**Transparent coating construction**

1-The laser cutter in the Fab lab is used to cut and shape the glassy Acrylic sheet into 6 faces. As appeared in **Figure 7,** the faces of the transparent prism –the external structure of the prototype– are classified into two 18.6×12 $cm^2$ rectangles acting as the bases of the prototype, one of them contains cut spaces for the LCD, switch button, and the Photovoltaic system—the Display face. Furthermore, two 18×7 $cm^2$ rectangles without cuts, which are the lateral faces; two 12×7 $cm^2$ rectangles, where one has sites to place the laser module and the laser diode—emission face.

2-The breadboard with all the connected wires is stabilized on the no cut 18.6×12 $cm^2$base

3-A glass/plastic adhesive is utilized to glue the complementary faces of the coating, thereby building the device. Moreover, the adhesive is used to attach the meter's components in their estimated placements within the coating faces.



*Figure 7 shows the project acrylic faces cut by the laser cutter*

**Test plan**

-The efficiency of the circuit code was verified by Arduino IDE before uploading it to the microcontroller. This is to prevent any errors arising during the process.

-The circuit connections and wiring between the components were effectively performed by a soldering pen—taking safety precautions into account. Therefore, a complete closed circuit is guaranteed.

-The power output from the suppliers was evaluated by a multimeter; thus, ensuring the power affordability for the device.

-The three dimensions for the prototype –length, width, and height– were measured by a typical tape. This is to find the volume of the instrument, so testing the portability.

-The ultrasonic sensor, a standard rangefinder, was used in the calibration of distance measurements with the Opti-Hawk meter and the motion sensor. This is to test obeying the maximum 5% error limit by the prototype in addition to comparing its efficiency with the motion sensor borrowed from the mechanics lab.

# Results

**Negative results**

- Due to extracting process mistakes, one of the pins in the reused ATmega8A is broken down, as in **Figure 8.** This arose a negative obstacle while constructing since D8 pin is connected to the Bluetooth module. Thus, it is solved finding an alternative input pin for the module in this microcontroller.
- Due to energy loss in the circuit's components, there is energy –in the form of heat energy– that is generated while the project's functionality, especially from the voltage regulator. Hence, a practical solution is to cut the external coating by the cutter in a manner by which an aeration opening is formed in the prototype's corners, as shown in **Figure 9**



*Figure 8 shows the aeration opening in Opti-Hawk meter*



*Figure 9 shows the broken PB8 (D8) pin of the ATmega*

**Positive results**

- Having overcome the negative results, the error percentage of Opti-Hawk meter was measured through calibration with a standard ultrasonic sensor connected to Arduino as well as the motion sensor from the Mechanics lab. Hence, the calculated length variations of the same moving object are illustrated in **Table 1**, plotted in **Figure 10,** to find the 2.7% average error of Opti-Hawk.
- The Opti-Hawk meter was used to test the lateral deflection (the strain) for one grade 10 project, the data is collected and graphed in **Table 2**, and **Figure 11**. This is to guarantee the applicability of the Opti-Hawk device in capstone test plans. Therefore, when the lateral pressure is applied, the prototype experienced a deflection of about 5-14 mm.
- The total volume of the Opti-Hawk prototype is calculated to be 0.0015 m$^2$; hence, the project obeys the estimated design requirement of portability (maximum 0.002 m$^2$ ).

*Table 1 shows the distance variations of a moving object*

| Sensor type | Ultrasonic | Opti-Hawk | Motion sensor |
|---|---|---|---|
| 0 sec | 297 ±1 | 303 ±1 | 317 ±1 |
| 1 sec | 290 ±1 | 287 ±1 | 289 ±1 |
| 2 sec | 272 ±1 | 267 ±1 | 268 ±1 |
| 3 sec | 245 ±1 | 255 ±1 | 249 ±1 |
| 4 sec | 235 ±1 | 245 ±1 | 233 ±1 |
| 5 sec | 229 ±1 | 235 ±1 | 227 ±1 |
| 6 sec | 214 ±1 | 218 ±1 | 213 ±1 |
| 7 sec | 195 ±1 | 196 ±1 | 205 ±1 |
| 8 sec | 180 ±1 | 185 ±1 | 189 ±1 |
| 9 sec | 167 ±1 | 171 ±1 | 193 ±1 |
| 10 sec | 150 ±1 | 161 ±1 | 193 ±1 |

*Table 2 shows the deflection (strain) of a grade 10 project*

| Distance | 199 mm | 210 mm | 214 mm | 208 mm | 195 mm |
|---|---|---|---|---|---|
| Time | 0 sec | 2 sec | 4 sec | 6 sec | 8 sec |
| | | Applying lateral pressure | | | |

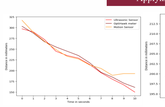



*Figure 10 shows the length variations by Opti-Hawk meter and other devices*

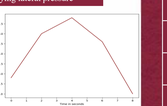



*Figure 11 shows a plot of the deflection as time passes*

# Analysis

**Laser module mechanism**

VL53L0X is the name of the rangefinder sensor for the project with a suitable range of 3-200 cm, measuring distances with an error of ± 1mm; thus, empowering its applications for accurate small length variations. For illustration, VL53L0X calculates distances by detecting the delay of reflected 940nm infrared (IR) pulses emitted by the emitter part; as a complement, **Figure 12** illustrates how the reflected beams are detected by a receiver part. Specifically, the pulses emission is the function of the Vertical Cavity Surface-Emitting Laser (VCSEL). This VCSEL is accompanied by a receiver for these reflected IR pulses, which is known as a Single Photon Avalanche Diode (SPAD). These components in the VL53L0X, the emitter and receiver array, are packaged in the module's integrated circuit (IC), as demonstrated in **Figure 13.** Furthermore, the left part of the figure clarifies that the VCSEL emits the IR beams in a 35◦ cone, while the multiple SPADs (SPAD array) perform in the collection exclusion cone of 25◦ ***(Chung, 2023).*** Consequently, this receiver sends instant signals, calculating the distance by a direct technique called Time-of-Flight (ToF). In fact, the length measured by the ToF technique is yielded from ***Equation 1 (Saad, 2024)***

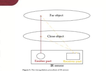



*Figure 12 shows shows a simple drawing for the IR rangefinder sensors, as VL53L0X* ***(Komarizadehasl et al., 2022)***

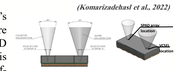



*Figure 13 shows a 3D demonstrative diagram for the VL53L0X IC* ***(Polulu, 2018)***

$$Distance = \frac{c \times \Delta t}{2} \quad (1)$$

Where $c$ is the constant speed of light in air, $\Delta t$ is the time taken by the reflected IR pulse.

Therefore, the VL53L0X model uses direct ToF to evaluate distances via its built-in mechanism; then, the calculated distances are transferred from the sensor to the Atmega8A in the form of precise voltage values through their communication protocol inter-Integrated Circuit (I$^2$C) ***(Polulu, 2018)***. Ultimately, the Atmega8A processes those voltages into the centimeter range form.

**Single Photon Avalanche Diode (SPAD)**

SPADs are photodiode detectors utilized in the project's laser module mechanism. It is called single photon since it is sensitive enough to sense only a photon hitting the active region; once arrives, the photon creates one electron, therefore stimulating the static electrons built up to initiate a serial chain resulting to a detectable digital signal—an avalanche. To clarify, the SPADs in VL53L0X are used in a 2D configuration in which many photodiodes are placed to form one polarized detector (SPAD array) ***(Saad, 2024).*** This polarization in SPAD is thanks to the dipole moment made by the occurrence of both cathode and anode, as shown in **Figure 14**. Hence, according to **(PH.2.02),** the dipole charging system creates an electric field that moves in the direction from the positive pole –the anode– to the negative one. –the cathode– which is reflected on the avalanche direction. For simplicity, the higher bias voltage on the terminals of the diode, the higher electric field intensity generated, the faster and stronger the movement of electrons in the avalanche; thus, increasing the observing efficiency, as the time resolution decreased. Clearly, the time resolution refers to how accurately the SPAD can tell when the photon arrived—0.1 nanosecond in the laser module situation ***(Giulia Acconcia et al., 2016)***. However, an external active restricting circuit is required to be applied on the SPAD to properly operate its mechanism—Active Quenching circuit. In other words, this external circuit quenches (stops) the avalanche signal produced by a photon to create another avalanche by the consequent photon—maintaining the continuous receiving signaling. Throughout this process, the laser module's SPAD detects 80 million photons per second—maximum count rate of 80 million counts/s. In fact, the maximum count rate is inversely proportional to the dead time after a detection event when the SPAD is unable to detect another photon, as explained in ***Equation 2 (Piron & Redouté, n.d.)***.

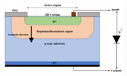



*Figure 14 shows a comprehensive cross-section for the structure of this SPAD*

$$DT = \frac{1}{PDR_{max}} \quad (2)$$

Where DT is the dead time of the SPAD detections, and $PDR_{max}$ is the maximum photon detection rate by SPAD. After substitution, the DT resulted in a value of 12.5 ns.

In conclusion, the value of the time resolution is essential in determining the overall accuracy of the laser module. Moreover, the relatively moderate count rate and the balanced rate of dead time by the SPAD array are certainly reflected in the results, especially in the quick data analysis.

**Electric Parameters of The Circuit**

Some of the important characteristics of the circuit that must be measured are the voltage, maximum drag electric current, and maximum operating power. A digital multimeter is used in measuring voltage and maximum drag electric current, where the reading of the voltage is 4.98 Volts, and that of the current is 0.55 Amperes. As studied in **(PH 2.03)**, the multimeter is connected in parallel to the battery when measuring voltage, while being connected in series during measuring current. Thereby, the maximum operating power is calculated by multiplying the voltage and maximum drag electric current ***(Sufyan, 2025).*** After substituting ***Equation 3***, the power is worked out to be 2.739 watts.

$$P_{max} = V_T I_{max} \quad (3)$$

To correlate with the resulting consumption, the co-power supplied by the photovoltaic cell is obtained. Firstly, the efficiency of the photovoltaic cells is calculated by ***Equation 4 (Wang et al., 2023).***

$$\eta_{PV} = \frac{MPP}{E \times A} \times 100 \quad (4)$$

Where MPP is the maximum power point, which is obtained experimentally as in ***Equation 3***. $E$ is the irradiance (intensity); it's the power of light incident per m$^2$ and considered a constant in the standard test conditions (STC): Irradiance of 1000 W/m$^2$, a temperature of 298.15◦ K, and an air mass of 1.5 ***(Murillo-Soto & Meza-Benavides, 2022).*** $A$ is the surface area of the photovoltaic cells, which is 5.625 × 10$^{-4}$m$^2$ in the prototype.

For illustration, the multiplication of the irradiance and the area yield the power applied on the photovoltaic cell, and it is calculated to give a value 0.5625. Thus, the efficiency of the photovoltaic cells is 3.68%, which is 1.32% less than the average for thin PV cells, which ranges from 5 to 7% ***(Al-Ezzi & Ansari, 2022).***

After calculating the efficiency of the photovoltaic cell, the alternative power provided to the battery can be obtained by identifying the circuit parts, which are: the PV cells, the charging module that receives up to 5 volts, and the battery. Hence, ***Equations 5,6 (Coridan et al., 2015)*** are used to get the power input to the battery by the PV cell.

$$P_{battery} = P_{PV} \times \eta_{total} \quad (5)$$

$$P_{battery} = P_{PV} \times (\eta_{charging\ module} \times \eta_{PV} \times \eta_{battery}) \quad (6)$$

The efficiency of the charging module ($\eta_{charging\ module}$) is 1. The reason is that the voltage generated by the photovoltaic cell cannot reach more than 5 Volts, so there will not be any heat loss when passing through the charging module. After plugging in with values, the net power that is provided is 0.0176 watts, which is considered a nearly sustainable suppliant to the battery.

**Quick Data Analysis**

In addition to displaying distance value on the LCD screen, data of measured distances is transmitted to a device such as the laptop via a Bluetooth module connected to the microcontroller. This data is saved in a CSV data file to use in drawing graphs, hence performing data analysis.

Clearly, the Bluetooth module is connected to two digital pins according to the circuit diagram (TX, RX) in ***(Figure 3)***. In fact, TX and RX pins are responsible for the serial communication between the module and microcontroller, where data travels from the TX of the module to the RX of the microcontroller, and vice versa. Due to the Universal Asynchronous Receiver/Transmitter (UART) connection, the baud rate - rate of transfer of data - of the module and the microcontroller must be the same to prevent time shift.

For simplicity, the data is transmitted across a virtual port, which is made possible by programming the microcontroller as in ***(Figure 4)***. Thereby, each time a distance is measured, its value is transmitted to the laptop via Bluetooth and saved in a CSV file, with the corresponding measurement time. As soon as the measurement stops, a graph is drawn using the saved data, as well as other visualizations using python code, in approximately 2 seconds.

Finally, the results are managed to be analyzed thanks to the learning outcomes of the curriculum shown in **Table 3:**

*Table 3 shows the learning transfer*

| Subject | Concepts | Connection |
|---|---|---|
| **CS.2.01** | Standard Network Topologies, and Types of Topologies | Visualizing and planning the circuit of the device as a network, and relying on the most suitable type of topology |
| **CS.2.02** | Databases concepts (Many-to-Many) | Creating effective databases involved in the prototype test planning—Bluetooth module library and libraries related to the quick data analysis. |
| **PH.2.05** | Capacitor charging and discharging | To achieve the full functionality of the device's microcontroller, two capacitors are connected in parallel to it |
| **PH.2.03** | Potential difference and voltage | Understanding all the voltage drops resulted from the circuit components; so, evaluating the proper operation power |
| **MA.2.01** | Polynomial graphing, End behaviour | While plotting the quickly collected measurements by the python code on graph, the behaviour of the graphed curve is identified, and determining the function when needed in the experiments |

# Conclusions

To conclude, the project succeeded in solving the problem of inaccurate distance measurements, measured by the motion sensor in the mechanics lab. The chosen solution resolved the drawbacks of prior solutions; notably the Bosch GLM20 laser meter situation, where the quick data analysis is performed by the project and graphs are visualized as well as displaying the distance measurement on an LCD. In fact, the graphs of quick data analysis were drawn after only 2 seconds from the end of the measurement. Further strong feature of the solution is that it uses an SPAD receiver, which improves time resolution to prevent time shift and loss of data, which has significant reduction of the 5% error maximum limit to 2.7%. This error was calculated after calibrating the device by an ultrasonic sensor, which is indeed an accurate rangefinder sensor—rather than the inefficient motion sensor. In addition, the project ensures sustainability in its structure, as it is composed of recycled components from old devices. Moreover, it uses renewable energy to power the device. To clarify, it uses photovoltaic cells to charge the lithium battery, by harnessing light energy. Although the results are promising, there are several proposed adjustments, which will be explained in the following section.

# Recommendation

**Real-life application**

In essence, the solution targets to address the problem of the inaccurate distance measurements and the absence of precision in the minute variations. Therefore, it's recommended to implement the project in the school's mechanics lab, shown in **Figure 15**, the site of curricular and capstone experiments. To elucidate, six experiments take place in the school's curriculum for grades 10 and 11, which are: thermal expansion, solving one-dimensional motion, projectile range, spring's constant, free fall acceleration, and strain.



*Figure 15 shows the location of STEM 6th of October important labs*

Strain is considered the most conducted experiment in the school for grade 10 students. In other words, the theme of the capstone project of grade 10 students is a construction theme, hence it clearly requires small distance variation measurements (strain). For example, grade 10 students in the last academic year had a test plan that required measuring the strain (deflection) of the deck of the bridge after putting some load on it. Currently, grade 10 students construct a residential building, a hospital, or a school. Hence, it's required to measure the lateral load on the building to ensure that the building withstands high wind speeds. Accordingly, the mechanics lab is considered the most suitable place for the device because of the continuous strain measurements that have the possibility to be repeated every year for grade 10 students.

**JRT laser range finder sensor**

**Figure 16** shows the powerful laser range finder module manufactured by JRT—a military grade distance sensor. This component specializes in the utilization of visible red laser beams in measuring –rather than the invisible infrared mostly used– although its compact size and its high portability to be used in devices; this is because it has dimensions of 45x25x12mm (±1 mm). Further features include the effective wide range of 8m, at the same time conserving the high accuracy of ±0.1 mm. In addition, JRT offers an extremely efficient time resolution for measurements (~0.125 seconds) and harsh environment endurance—applicable in temperatures of 0-40 ℃ ***(JRT measure, 2023)***.



*Figure 14 shows a clear image for JRT laser sensor*

This proficient sensor is favored to replace both the infrared laser module and the laser diode, thanks to all these functional advantages, as well as decreasing the complexity of the internal structure of the Opti-Hawk meter. Nonetheless, this recommendation was not applied in the project because this sensor is not only manufactured and sold with high costs in China, but also it is found separately, making it impossible to be recycled— which violates the design requirements.

**58mm thermal printer**

**Figure 17** shows the 58mm thermal printer, which is the smallest and most practical thermal printer. The device is considered ideal for representing graphs of results from various experiments, whether related to the curriculum or the capstone. This is because offline representation is stated to be better than online representation in a study, where the remembering rate of students of the content was 21% more than those who studied without experiments ***(Feverding, 2016).*** This device is compatible with simple circuits through the TTL outlet, providing an easy connection to the microcontroller. It consists of the driver board, on which the bitmap graphics (specific file extensions to overlap results with the graph) are uploaded with the aid of the TTL Outlet. In addition, it includes a power outlet to provide the printer with 5V from the microcontroller. However, this printer cannot be found in old devices and must be bought as new within the range of 25 dollars. So, it would be impossible to use it in the prototype.




*Figure 15 illustrates the structure and the parts of the 58mm thermal printer*

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